

Anthesis: Governed Execution for Agentic Systems

Ryan Jennings

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Anthesis Whitepaper

Version and authorship

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This publication release completes the integration ownership matrix, records the signed 24-case Governance Lab proof path, and binds the public PDF to reviewed source and artifact digests. Earlier revisions remain available through version control.

1. Executive summary

Anthesis is the governance boundary between an agent's intent and its externally observable effects.

It evaluates consequential actions against explicit policy, authority, approvals, capabilities, and evidence requirements before those actions cross a real-world boundary. The result is a structured decision that a surrounding tool, gateway, credential boundary, validator, or runtime can enforce.

Anthesis is designed to compose with existing agent frameworks and execution environments. It is not a general-purpose agent orchestrator, model runtime, memory database, project-management system, or compliance dashboard.

Its central invariant is:

Every governed externally observable effect is authorized, constrained, attributable, and auditable through Anthesis.

The strength of that claim depends on the integration preventing bypass. Prompts, conventions, and cooperative wrappers are useful ergonomics, but they are not enforcement guarantees by themselves.

2. The problem

Agentic systems can acquire practical authority through tool registries, credentials, network access, delegated specialists, filesystem access, and automated loops. Without a clear governance boundary, authority may be implicit, difficult to review, and hard to reconstruct after an incident.

A useful governance system must answer four questions before a consequential effect occurs:

1. **What is being requested?**
2. **Who or what has authority to request it?**
3. **Which policy, approval, capability, and evidence requirements apply?**
4. **What prevents execution outside the approved boundary?**

Anthesis addresses the decision and evidence parts of this boundary. Enforcement is completed by the selected integration: a restricted tool surface, gateway, downstream capability validator, credential boundary, sandbox, or combination of controls.

3. Governed effects

Reasoning may remain internal. Effects must cross a governance boundary.

Governed effects may include:

- specialist or sub-agent delegation;
- tool and MCP invocation;
- memory or context access;
- filesystem and repository mutation;
- shell or process execution;
- network and API calls;
- external communication;
- approval-gated actions;
- deployment, release, or administrative operations;
- artifact publication and durable state mutation.

Not every effect needs the same policy or approval level. Low-risk actions may be permitted under an existing policy grant. Higher-risk actions may require exact human approval, narrower capabilities, stronger evidence, or a safe halt.

4. Responsibility boundaries

Anthesis is part of a broader proof and execution ecosystem with deliberately separate responsibilities.

Component	Responsibility
Anthesis	Policy authority, deterministic decision contracts, approval requirements, capability constraints, evidence semantics, and provenance.
Anthesis Governance Lab	Independent deterministic conformance, scenario packs, aggregate reports, and public walkthroughs against the Anthesis evaluator contract.
Dubnium	Reproducible local-first agentic runtime, gateway, bounded tools, execution, and runtime evidence for a reference integration.

Governance Lab is not the deployable governance product and is not in the runtime critical path. It evaluates declared scenarios and does not execute their declared effects.

Dubnium is not the policy authority. It demonstrates how an execution environment can consume Anthesis decisions, enforce exact approvals, run bounded actions, and return evidence.

5. Governed effect flow

A typical governed action follows this shape:

```

Agent or supervisor request
-> normalize the proposed effect
-> Anthesis policy and authority evaluation
    -> allow with constraints
    -> exact approval required
    -> deny or safe halt
-> integration enforcement boundary
-> bounded execution
-> evidence and outcome return
-> audit, evaluation, and recovery as applicable

```

Five concerns must remain distinct:

Concern	Responsibility
Decision integration	Anthesis normalizes and evaluates a proposed externally observable effect.

Concern	Responsibility
Enforcement integration	A tool, gateway, credential boundary, downstream validator, or runtime prevents an unauthorized effect.
Evidence integration	Execution identity, material activity, outcome, and validation evidence return to the governed record.
Reference validation	Governance Lab independently validates public Anthesis contracts and deterministic decisions.
Reference execution	Dubnium demonstrates bounded execution using the public Anthesis decision boundary.

A policy decision without enforcement can still be bypassed. Enforcement without evidence may block some misuse but remain difficult to inspect or audit. A complete governed path needs all three operational concerns: decision, enforcement, and evidence.

6. Integration and enforcement model

Anthesis supports six canonical integration modes:

1. **Tool wrapper / invoke**
2. **MCP mediation**
3. **Gateway / sidecar**
4. **Capability tokens**
5. **SDK wrapper**
6. **Sandboxed runtime**

The modes describe integration mechanisms. They do not carry a fixed guarantee level by name alone.

6.1 Enforcement location

Enforcement may live at one or more locations:

- **Application** - an SDK or application wrapper controls how effects are requested.
- **Tool surface** - a tool registry or MCP facade controls the agent-facing capability surface.
- **Infrastructure** - a gateway, sidecar, credential boundary, network control, or downstream validator controls access to effects.
- **Runtime** - a sandbox controls filesystem, network, process, credential, and tool access.

6.2 Assurance classification

The resulting assurance depends on the surrounding bypass controls:

- **Advisory** - Anthesis recommends or records a decision, but direct effect paths remain available.
- **Moderate** - agents are configured to use Anthesis paths, but additional bypass paths may remain.
- **Strong** - registries, credentials, network controls, gateways, or downstream validators prevent most direct bypass.
- **Runtime-enforced** - the runtime prevents direct effects outside governed paths.

The same nominal mode may fall into different classifications. MCP mediation with raw credentials and unrestricted network access may be moderate. MCP mediation combined with credential isolation and egress controls may be strong.

6.3 Integration ownership matrix

The ownership matrix is presented as one readable contract per mode. Every contract covers:

- **Agent-facing surface and enforcement location;**
- **Authority, credentials, and executor;**
- **Required bypass prevention and evidence return;**
- **Typical assurance, trust assumptions, and recommended composition.**

Tool wrapper / invoke

Field	Contract
Agent surface	Restricted tool registry or invoke endpoint.
Enforcement location	Tool surface.
Authority and execution	Anthesis owns policy authority. A wrapper or broker owns credentials. A bounded tool performs the effect.
Bypass prevention	Raw tools and credentials must be unavailable to the agent.
Evidence return	Request digest, decision, approval or capability reference, executor identity, and outcome.
Assurance and trust	Moderate to strong. Trust the registry and wrapper to preserve action binding.
Recommended composition	Credential isolation and an evidence callback.

MCP mediation

Field	Contract
Agent surface	Anthesis-facing MCP facade.
Enforcement location	Tool surface.
Authority and execution	Anthesis owns policy authority. The mediator or broker owns downstream credentials. A registered downstream MCP server or tool executes.
Bypass prevention	Raw MCP servers, service credentials, and direct service network paths must be unavailable or constrained.
Evidence return	Tool identity, sanitized argument digest, decision reference, and result evidence.
Assurance and trust	Moderate to strong. Trust client configuration only when raw server registration and direct egress are controlled.
Recommended composition	Credential isolation and network egress control.

Gateway / sidecar

Field	Contract
Agent surface	Governed gateway endpoint.
Enforcement location	Infrastructure.
Authority and execution	Anthesis owns policy authority. The gateway or sidecar owns service credentials. The downstream service executes.
Bypass prevention	Network routes and service credentials must force traffic through the gateway.
Evidence return	Resolved route, action binding, decision, approval state, executor, and outcome.
Assurance and trust	Strong. Trust gateway availability, route integrity, and downstream identity.
Recommended composition	Capability validation and runtime isolation.

Capability tokens

Field	Contract
Agent surface	Action-bound capability envelope presented to a gateway, tool, or service.
Enforcement location	Infrastructure or downstream effect boundary.
Authority and execution	Anthesis policy and attributable approval grant the bounded capability. The downstream validator owns grant validation, and the executor performs the effect.
Bypass prevention	Missing, altered, expired, replayed, or out-of-scope grants must be rejected at the effect boundary.
Evidence return	Invocation receipt linked to the decision and result.
Assurance and trust	Strong. Trust integrity keys, audience binding, revocation, and downstream validation.
Recommended composition	Gateway, SDK, or sandbox.

SDK wrapper

Field	Contract
Agent surface	Application API or library call.
Enforcement location	Application.
Authority and execution	Anthesis owns policy authority. The application or a broker owns credentials. The application invokes the executor.
Bypass prevention	Direct clients and raw credentials must be removed or constrained; otherwise the wrapper is advisory.
Evidence return	Decision and execution evidence through the SDK callback.
Assurance and trust	Advisory to moderate. Trust the application process and its operators.
Recommended composition	Brokered credentials or a gateway for stronger assurance.

Sandboxed runtime

Field	Contract
Agent surface	Only sandbox-approved filesystem, network, process, credential, and tool surfaces.
Enforcement location	Runtime.
Authority and execution	Anthesis owns policy authority. The sandbox or broker owns credentials. A bounded runtime and governed external gateways execute.
Bypass prevention	The runtime must prevent direct effects outside governed paths and preserve execution identity.
Evidence return	Process, tool, network, artifact, and outcome evidence.
Assurance and trust	Runtime-enforced when complete. Trust the host, sandbox boundary, policy configuration, and escape resistance.
Recommended composition	Governed external gateway and scoped capabilities.

For every deployment, identify the exact direct-effect path being removed or constrained. The matrix does not make a mode production-grade by name; the surrounding controls determine assurance.

Detailed Questions, Alternatives, Recommendations, and Tradeoffs analyses remain in the product integration documentation rather than being duplicated here.

7. Composition patterns

Production and higher-assurance deployments will often combine modes.

SDK wrapper plus credential isolation

The SDK provides a simple application interface, while raw service credentials are held by a gateway or broker. The wrapper alone is not the guarantee; credential ownership and downstream access control prevent bypass.

MCP mediation plus egress control

The agent sees only the Anthesis MCP facade. Network policy prevents direct access to downstream services, and credentials are unavailable to the model-

facing process. This can raise an otherwise cooperative integration to strong assurance.

Gateway plus capability validation

Anthesis issues an action-bound capability after policy evaluation. The gateway and downstream service reject requests without a valid grant. Evidence links the decision, capability, executor, and outcome.

Sandboxed runtime plus gateway and scoped capabilities

A sandbox prevents direct filesystem, network, process, and credential access. External effects pass through a governed gateway and require scoped grants. This provides the strongest reference pattern, with greater operational complexity.

Tool wrapper plus evidence callback

For a low-risk local trial, a restricted tool registry may expose only Anthesis-scoped actions and return material evidence after execution. This is useful for adoption testing but should not be represented as universally non-bypassable without environmental controls.

8. Public proof paths

8.1 Runnable now: Governance Lab

The public evaluator and Anthesis Governance Lab provide an independently runnable deterministic proof surface:

- signed immutable evaluator acquisition with Sigstore verification before checksum, provenance, extraction, or execution;
- seven canonical public-contract scenarios;
- nine general demonstration packs containing 27 synthetic scenarios;
- 24 provider-neutral inference-integrity scenarios with deterministic JSON and YAML reports;
- controlled expectation drift returning stable exit code 7;
- reproducible evidence bundles and stakeholder walkthroughs;
- no requirement for a GPU, live model, private source repository, production credentials, or hosted service.

Governance Lab proves public contract and recorded-evidence behavior. It does not execute declared file, network, deployment, merge, release, or administrative effects, and it does not prove production non-bypassability by itself.

8.2 Reference integration: Dubnium

Dubnium demonstrates a bounded governed-agent execution path:

Human task

- > exact structured action request
- > Anthesis decision
- > action-bound approval when required
- > bounded Dubnium executor
- > visible outcome and sanitized evidence

The reference integration consumes the same signed immutable Anthesis evaluator identity used by Governance Lab. This is a reference composition showing decision, approval, execution, and evidence responsibilities. It is not a claim that every Anthesis deployment has equivalent runtime enforcement.

8.3 In development

Active work includes broader production enforcement profiles and stronger live runtime evidence, protected replay, independent verification, containment, recovery, and production inference-integrity controls.

The runnable 24-case suite evaluates recorded provider-neutral evidence. It must not be described as complete live prevention of model-weight or covert-channel exfiltration.

9. Security and trust boundaries

Anthesis is fail-closed at the decision boundary when required policy, identity, approval, capability, or evidence inputs are missing or invalid. Deployment assurance still depends on the enforcing environment.

A security review should test at least these conditions:

- direct tool or API invocation without Anthesis;
- use of raw credentials outside the governed path;
- expired, altered, replayed, or out-of-scope capabilities;
- approval reuse against a different action, target, or payload;
- direct network, filesystem, process, or repository effects;
- missing or contradictory execution identity;
- evidence mutation or replacement after execution;
- fallback to an unregistered runtime, model route, verifier, or tool;
- failure to return material outcome and validation evidence.

The system should state residual bypass assumptions explicitly. A prompt instruction to “always call Anthesis first” is not equivalent to registry restriction, credential isolation, downstream validation, or sandbox enforcement.

10. Adoption path

A practical evaluation can proceed in three stages:

1. **Validate the contract.** Run Governance Lab and inspect deterministic decisions, reasons, rules, and reports.
2. **Integrate one bounded effect.** Select a low-risk action, identify its bypass paths, choose an integration mode, and require evidence return.
3. **Strengthen enforcement.** Add credential isolation, gateway controls, downstream capability validation, egress restriction, or sandboxing according to the risk.

A trial is successful when an evaluator can explain:

- what Anthesis decided and why;
- what prevented the effect from bypassing the decision;
- who or what executed the action;
- which approval or capability authorized it;
- what evidence proves the material outcome;
- what trust assumptions and residual risks remain.

11. Publication integrity

The public artifact is generated only from the explicit whitepaper source manifest. The complete Anthesis documentation corpus, when built, uses the distinct name `anthesis-documentation.pdf` and is not published as the whitepaper.

Each reviewed whitepaper build emits a trace manifest containing:

- the Anthesis source commit;
- the curated source manifest and its digest;
- every source and build-input digest;
- the document metadata contract;
- the generated PDF SHA-256 digest.

The community publication path commits the PDF and trace manifest together. Community CI rejects a missing, malformed, or digest-mismatched pair before deployment.

12. Scope and non-goals

Anthesis does not automatically make an agentic system secure merely by being called. It does not guarantee deterministic model output, universal replay, complete compliance, or non-bypassability without an enforcing integration.

The public whitepaper is explanatory. Accepted RFCs, schemas, policy contracts, and evaluator contracts remain the normative sources for implementation behavior.

13. Conclusion

Anthesis makes agent authority explicit at the point where intent becomes effect.

Its value is not only that it can return an allow, approval-required, or deny decision. Its value is that existing agentic systems can bind those decisions to enforcement controls and evidence records that remain reviewable after execution.

The governing question for every integration is therefore: **What prevents the agent from producing this effect without crossing Anthesis?**